Logistic Regression

In **regression problems**: Y is **quantitative**.

In **classification problems**: Y is **qualitative**.

Qualitative variables take values in an unordered set $\mathcal{C}$, for example:

1. hand-written number $\in \{0, 1, \ldots 9\}$.
2. transaction $\in \{normal, fraudulent\}$.

Classification task: given a feature vector X and a qualitative response $Y \in \mathcal{C}$, build a function $C(X)$ (a.k.a. a classifier) that takes X as input and predicts the value for Y.

Often we are more interested in estimating the **probabilities** that Y belongs to each category in $\mathcal{C}$.

Classification method -- Logistic Regression

1. Why not Linear Regression?

Multi-class classification problem:
Consider a classification problem in which we want to classify a set of bonds into three categories: $C = \{A, B, C\}$. It is natural to encoding Y as:

$$Y = \begin{cases} 1, & if\ A \\ 2, & if\ B \\ 3, & if\ C \end{cases}$$

Problem: These classes are not ordered, but these labels are.

How about **Binary classification problem**?
Consider the $Default$ data as an example, we can encode Y as:

$$Y = \begin{cases} 0, & if\ no \\ 1, & if\ yes \end{cases}$$

Can we simply perform a linear regression of Y on X and classify as Yes if $\hat{Y} > 0.5$? This question can be expressed as follows:

Question: What is our regression function estimating?

In this case, $E(Y|X=x) = Pr(Y=1|X=x)$, so the regression function is actually estimating the probability that $Y=1$ given $X=x$, looks great! **But it now works**.

Actually, linear regression uses predictor $\boldsymbol{x}\hat{\boldsymbol{\beta}}$ to predict $E(Y|X=\boldsymbol{x}) = Pr(Y=1|X=\boldsymbol{x})$.

Hence, we predict $\hat{y}=1$ if $\boldsymbol{x}\hat{\boldsymbol{\beta}} > 0.5$ else $\hat{y}=0$.

Summary:
In general, regression methods are not suitable for classification problems for at least two reasons:

1. a regression method cannot accommodate a qualitative response with more than two classes;
2. a regression method will not provide meaningful estimates of $Pr(Y|X)$, even with just two classes.

2. Generalized Linear Models (GLM)

A new look at (ordinary) linear regression:

$$\boldsymbol{y} = \boldsymbol{X}\boldsymbol{\beta} + \boldsymbol{\epsilon}, \boldsymbol{\epsilon} \sim N(\boldsymbol{0}, \sigma^2\boldsymbol{1})$$

If we denote $E(\boldsymbol{y}|\boldsymbol{X}) = \boldsymbol{\mu}$, then we have:

- $\boldsymbol{y} \sim N(\boldsymbol{\mu}, \sigma^2\boldsymbol{I})$, a distribution for the response variable
- $\boldsymbol{\eta} = \boldsymbol{X}\boldsymbol{\beta}$, a linear predictor
- $\boldsymbol{\mu} = \boldsymbol{\eta}$, identity link function

Main problem for the **Default** data set: Normal distribution is **NOT appropriate** for modeling the response variable taking 0 − 1 values.

Generalized linear models (GLM) extends the (ordinary) linear regression models by allowing for a variety of distributions for $\boldsymbol{y}$.

All GLM's have the following three key components:

- A probability distribution (with some parameter θ) describing the response variable y;
- A linear predictor $\eta = \beta_0 + \beta_1 x_1 + \cdots + \beta_p x_p$;
- A link function $g()$ that relates the linear model to the parameter of the outcome distribution:

$$g(\theta) = \eta$$

Model	Distribution of Y	Support of distribution	Key parameter	Link function	Typical uses
OLR	$N(\mu, \sigma^2)$	Real: $(-\infty, \infty)$	μ	Identity link: $g(\mu) = \mu$	Linear-response data
Logistic regression	$Bernoulli(p)$	Integer: $\{0, 1\}$	p	Logit link: $g(p) = log(\frac{p}{1-p})$	Binary classification
Poisson regression	$Poisson(\mu)$	Integer: $\{0, 1, 2, \cdots\}$	μ	Log link: $g(\mu) = log(\mu)$	Count of occurrences in fixed amount of time/space

3. Logistic Regression

3.1 Model Specification

A logistic regression model is typically specified as follows:

$$\begin{aligned} Y_i &\sim Bernouli(\pi_i) \\ logit(\pi_i) &= \eta_i \\ \eta_i &= \beta_0 + \beta_1 x_{i1} + \cdots + \beta_p x_{ip} \end{aligned}$$

Where:

- $\pi_i = P(Y_i = 1) = E(Y_i)$
- The **logit link function** is $logit(\pi) = log(\frac{\pi}{1-\pi})$
- By inverting the logit function, we have

$$\pi_i = \frac{exp(\eta_i)}{1 + exp(\eta_i)}$$

3.2 Estimation of Model Parameters

The ordinary least squares (OLS) method is no longer capable of estimating the logistic regression model.

The logistic regression is usually estimated by the **maximum likelihood estimation (MLE)** method.

In general, if a random sample $X_1, \cdots, X_n \overset{i.i.d.}{\sim} f(x|\theta)$, then the likelihood function (of θ, given $X = x$) is defined by:

$$L(\theta|x) = \prod_{i=1}^{n} f(x_i|\theta)$$

The MLE of θ is defined by:

$$\hat{\theta} = arg \max_{\theta} L(\theta|x)$$

For logistic regression, the MLE of $\boldsymbol{\beta}$ seeks for the vector $\boldsymbol{\beta} = (\beta_0, \cdots, \beta_p)$ that maximizes the log-likelihood function:

$$\mathcal{L}(\boldsymbol{\beta}|\boldsymbol{y}, \boldsymbol{X}) = \sum_{i=1}^{n} \left[y_i \boldsymbol{x}_i^T \boldsymbol{\beta} - log(1 + e^{\boldsymbol{x}_i^T \boldsymbol{\beta}}) \right]$$

This optimization problem does not have a closed-form solution, it typically requires a **numerical optimization procedure**.

3.3 Feature Selection

Classic method: stepwise regression + AIC/BIC

$$AIC = -2\text{log-likelihood} + 2(p+1)$$
$$BIC = -2\text{log-likelihood} + log(n)(p+1)$$

Shrinkage estimation: LASSO, Adaptive LASSO, SCAD, MCP, ...

3.4 Predicting Classification Assignment

Suppose that the estimated coefficient is $\hat{\boldsymbol{\beta}} = (\beta_0, \cdots, \beta_p)^T$, then the predicted probability for $Y = 1$ at a new observation x^* is:

$$\hat{\pi}(x^*) = \frac{exp(x^* \hat{\boldsymbol{\beta}})}{1 + exp(x^* \hat{\boldsymbol{\beta}})}$$

This gives the classifier:

$$\hat{Y} = \hat{C}(x^*) = \begin{cases} 1, & \hat{\pi} > c \\ 0, & \hat{\pi} \leq c \end{cases}$$

where c is the cut-off value, typically we take $c = 0.5$.

4. Evaluation of Classification Accuracy

4.1 Condusion Table

		Predicted condition	
	Total population = P + N	Positive (PP)	Negative (PN)
Actual condition	Positive (P)	True positive (TP), hit	False negative (FN), type II error, miss, underestimation
	Negative (N)	False positive (FP), type I error, false alarm, overestimation	True negative (TN), correct rejection

- Sensitivity (Racall, True Positive Rate): $TPR = \frac{TP}{TP+FN}$;
- Specificity (True Negative Rate): $TNR = \frac{TN}{TN+FP}$;
- Precision (Positive Predictive Value): $PPV = \frac{TP}{TP+FP}$;
- False Positive Rate: $FPR = \frac{FP}{FP+TN} = 1 - TNR$;
- False Discovery Rate (FDR): $FDR = \frac{FP}{FP+TP} = 1 - PPV$;
- Classification Accuracy: $ACC = \frac{TP+TN}{P+N}$.

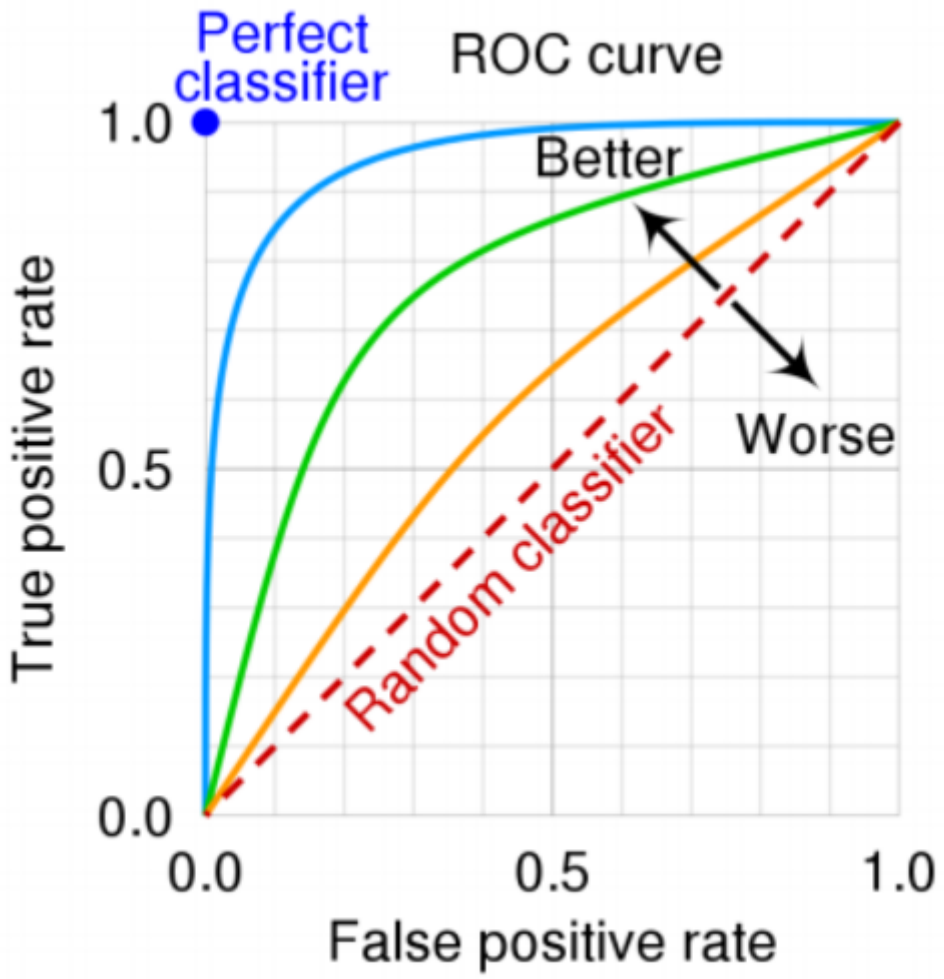
4.2 ROC and AUC

Recall that the classification results depends on the cut-off value c, which is typically taken as 0.5.

By changing the value of c we obtain a sequence of classifiers with their own sensitivity and specificity.

By plotting $sensitivity$ against $1 - specificity$, we obtain the **Receiver operating characteristic (ROC)** curve.

Area under the ROC curve is called AUC, which is an overall measure of the predictive performance of the model. **Larger AUC, better model**.


Perfect classifier
ROC curve
1.0
Better
Worse
0.5
Random classifier
0.0
True positive rate
0.0
0.5
1.0
False positive rate